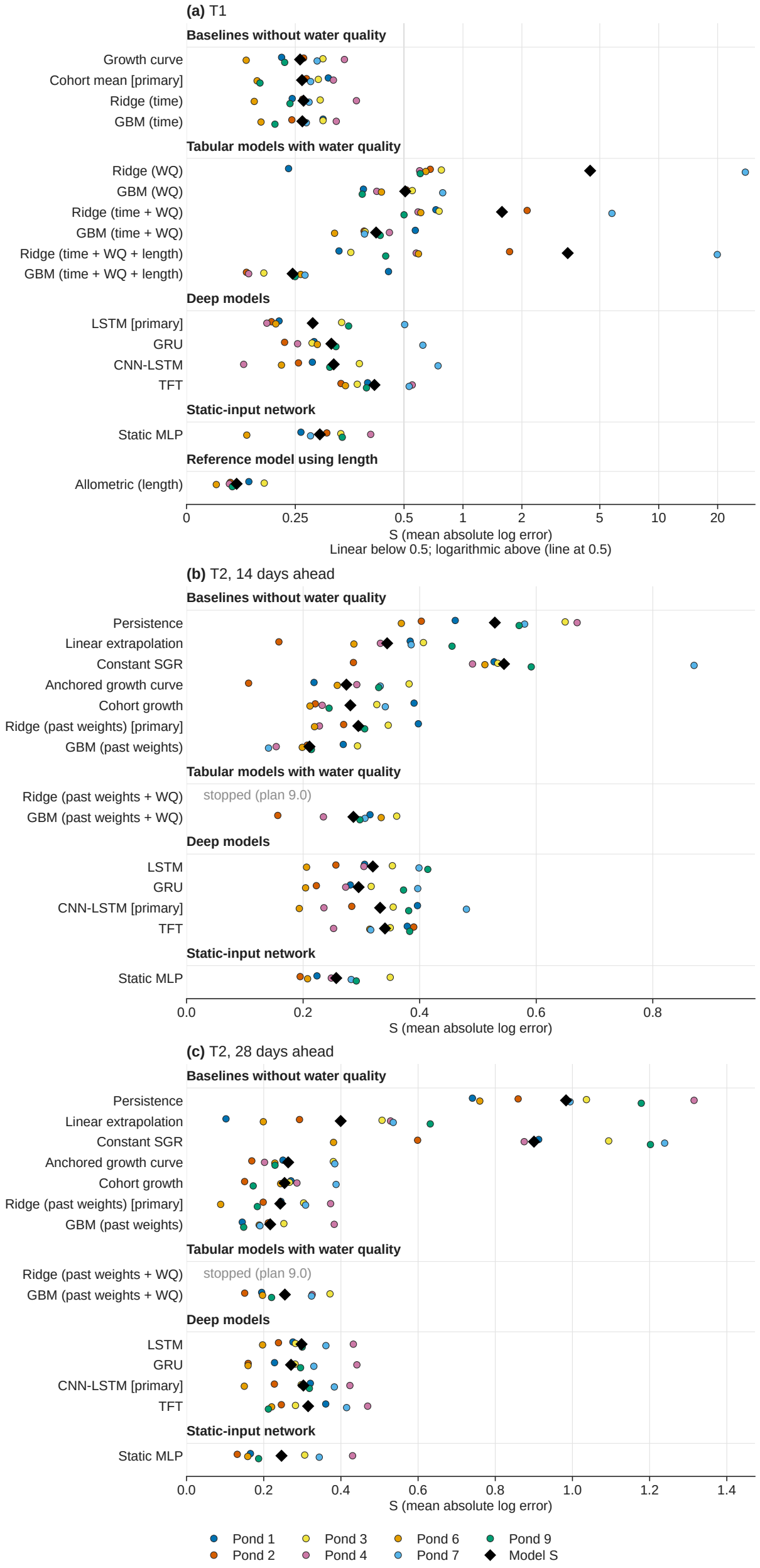

(a) T1
Baselines without water quality
Growth curve
Cohort mean [primary]
Ridge (time)
GBM (time)
Tabular models with water quality
Ridge (WQ)
GBM (WQ)
Ridge (time + WQ)
GBM (time + WQ)
Ridge (time + WQ + length)
GBM (time + WQ + length)
Deep models
LSTM [primary]
GRU
CNN-LSTM
TFT
Static-input network
Static MLP
Reference model using length
Allometric (length)
0
0.25
0.5
1
2
5
10
20
S (mean absolute log error)
Linear below 0.5; logarithmic above (line at 0.5)
(b) T2, 14 days ahead
Baselines without water quality
Persistence
Linear extrapolation
Constant SGR
Anchored growth curve
Cohort growth
Ridge (past weights) [primary]
GBM (past weights)
Tabular models with water quality
Ridge (past weights + WQ)
stopped (plan 9.0)
GBM (past weights + WQ)
Deep models
LSTM
GRU
CNN-LSTM [primary]
TFT
Static-input network
Static MLP
0.0
0.2
0.4
0.6
0.8
S (mean absolute log error)
(c) T2, 28 days ahead
Baselines without water quality
Persistence
Linear extrapolation
Constant SGR
Anchored growth curve
Cohort growth
Ridge (past weights) [primary]
GBM (past weights)
Tabular models with water quality
Ridge (past weights + WQ)
stopped (plan 9.0)
GBM (past weights + WQ)
Deep models
LSTM
GRU
CNN-LSTM [primary]
TFT
Static-input network
Static MLP
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
S (mean absolute log error)
Pond 1
Pond 2
Pond 3
Pond 4
Pond 6
Pond 7
Pond 9
Model S